Challenge Brief

Template for challenge partners · Since AI Hackathon 2026 · 6–8 Nov 2026, Turku

This template helps you define a clear, buildable challenge for the hackathon. Fill in each section below — the more concrete the problem and the evaluation criteria, the better the prototypes you receive. You can return it to us as is, and we are happy to help shape it into a ready event challenge.

Questions about the template? Riku Lauttia · riku.lauttia@sinceai.fi · +358 40 938 8820

Challenge partner (company)

Business Turku Oy Ab
Technical sparring: Elomatic.

01 Challenge Title

A short, memorable name for the challenge. Example: “Automated support ticket routing.”

RETROFIT AI — Make Existing Machines Intelligent
Give existing industrial equipment predictive intelligence without replacing its machinery or control system.

02 Problem Statement

What business or technical problem should be solved? Describe the context, why it matters, and the main pain points.

Existing pumps and motors on ships and in factories often lack predictive monitoring. Operators may detect wear late, while replacing functional equipment or its controls to add AI is costly—especially for SMEs.
Build an affordable intelligence layer using available sensor signals. Demonstrate early fault detection, understandable evidence and operator recommendations on a simulated cooling-water pump. Explain how the approach could transfer to other rotating machines.
Keep safety controls unchanged; any proposed operating adjustment is tested only in the simulator, never on live equipment.

03 Guiding Question

One question that focuses the challenge. Example: “How might we use AI to predict equipment failures before they happen?”

How might we make existing industrial machines more reliable, efficient and understandable through low-cost sensing and edge-capable AI—without replacing the machines or their safety-critical controls?

04 Recommended AI Approaches *Optional*

Any relevant technologies, models or methods — for example NLP for text analysis, classification, knowledge graphs, agents, or RAG.

Possible methods: time-series anomaly detection, fault classification, early-warning models, physics-informed features, uncertainty-aware diagnostics, edge inference and simulator-based what-if testing.
Teams may demonstrate a read-only OPC UA/MQTT-style adapter or a natural-language explanation interface. Neither is required. A chatbot alone is insufficient: the core must be a measurable, functioning industrial intelligence workflow.

05 Dataset or API Provided *If any*

Describe any data or resources you provide: format, a link if ready, and sample size. If none, state what builders should expect.

> At Since AI, challenges can be built with synthetic, anonymized or public data — no personal data is required.

The challenge provides a ready-to-use, fully synthetic pump benchmark. It contains 12 simulated ship and industrial cooling/process pumps, seven days of telemetry per pump at 1 Hz (about 7.2 million rows), and a smaller 10-second starter set. It includes no real measurements from Elomatic, vessels, plants or customers.
Download the included RETROFIT_AI_PARTICIPANTS_2026.zip: telemetry (Parquet / CSV.gz), operating context, alarms, maintenance logs, sensor features, vibration snapshots, data dictionary, documented simulator, what-if sandbox, example code and baselines. Train/validation events are labelled; test-event labels are held back for judging.
Start with START_HERE.html and README_FIRST.md inside the data ZIP. No equipment, external API or confidential industrial dataset is required. All machine-control actions must stay in the simulator.

06 Evaluation Criteria

How should solutions be judged? What matters most — for example accuracy, scalability, usability, or business impact?

A. Operational value and decision usefulness — 25%: actionable alerts, clear explanations and credible benefits for maintenance or energy use.
B. Technical validity — 25%: measured detection quality and warning lead time; comparison with a simple baseline; reproducible results.
C. Reliability and safety — 20%: false alarms, missing sensors, uncertainty and safe human review; no unverified live actuation.
D. Retrofit feasibility and transferability — 20%: plausible sensor/integration cost, edge or on-site operation, and adaptation to another machine class.
E. Working prototype and communication — 10%: usable end-to-end demonstration and clear limitations.
TOTAL: 100%.

07 Submission Format

What deliverables do you expect? For example a demo video, pitch slides, a code repository, or documentation.

Submit through sinceai.app by the official event deadline. Demonstrate a working operator-assist prototype on the supplied synthetic data: normal operation, developing faults, evidence behind alerts, uncertainty and safe, human-reviewable actions. No live machine control.
Provide a short narrated demo video and concise PDF/presentation, source code with reproducible run instructions, a baseline/metrics comparison and a technical note on retrofit costs, deployment assumptions, safety and limits. Include a test-pump alert CSV matching submission_template/predictions_template.csv from the data pack.
Optional: simulation-only what-if comparison or credible transfer to another rotating machine. Official Since AI submission and intellectual-property rules apply.

08 Point of Contact

Name, email and preferred contact method for questions during the hackathon.

Challenge owner and contact: Reetta Vuorinen, Business Turku — reetta.vuorinen@businessturku.fi.
Technical sparring: Joonas Riekkinen, Elomatic — joonas.riekkinen@elomatic.com (onsite Friday; subsequent availability may vary).
Event coordination: Riku Lauttia, Since AI ry — riku.lauttia@sinceai.fi. Please use the official Since AI Discord for builder questions; organizer staff will coordinate escalation and evaluation access.